

Model Optimization and Tuning Phase Template

Date	20-02-2026
Team ID	LTVIP2026TMIDS62277
Project Title	Prosperity Prognosticator: Machine Learning for startup success prediction
Maximum Marks	10 Marks

Model Optimization and Tuning Phase

The Model Optimization and Tuning Phase involves refining neural network models for peak performance. It includes optimized model code, fine-tuning hyperparameters, comparing performance metrics, and justifying the final model selection for enhanced predictive accuracy and efficiency.

Hyperparameter Tuning Documentation (8 Marks):

Model	Tuned Hyperparameters

Gradient

Boosting

Classifier

#importing the library for grid search

from sklearn.model_selection import GridSearchCV

The 'lr_param_grid' specifies different values for regularization strength (C), solvers (solver), and penalty types (penalty). GridSearchCV (lr_cv) is employed with 5-fold cross-validation (cv=5), evaluating model performance based on accuracy (scoring="accuracy"). The process uses all available CPU cores (n_jobs=-1) for parallel processing and provides verbose output (verbose=True) to track progress.

```
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score
from sklearn.ensemble import GradientBoostingClassifier
rf = GradientBoostingClassifier()
param_grid = {'n_estimators': [100, 200, 300], 'criterion': ['entropy', 'gini'], 'max_depth': [10, 20, 30], 'max_features': ['auto', 'best']}
grid_search = GridSearchCV(rf, param_grid, cv=5, n_jobs=-1, verbose=0)
grid_search.fit(X_train, y_train)

✓ 11s

Fitting 5 folds for each of 16 candidates, totalling 80 fits

+ GridSearchCV
- best_estimator_ : GradientBoostingClassifier
  - GradientBoostingClassifier
```

Random Forest

The parameter grid (rfc_param_grid) for hyperparameter tuning. It specifies different values for the number of trees (n_estimators), splitting criterion (criterion), maximum depth of trees (max_depth), and maximum number of features considered for splitting (max_features). GridSearchCV (rfc_cv) is employed with 3-fold cross-validation (cv=3), evaluating model performance based on accuracy (scoring="accuracy").

```
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier()
param_grid = {'n_estimators': [100, 200, 300], 'criterion': ['entropy', 'gini'], 'max_depth': [10, 20, 30], 'max_features': ['auto', 'best']}
grid_search = GridSearchCV(rf, param_grid, cv=3, n_jobs=-1, verbose=0)
grid_search.fit(X_train, y_train)

✓ 4s

Fitting 3 folds for each of 16 candidates, totalling 48 fits

+ GridSearchCV
- best_estimator_ : RandomForestClassifier
  - RandomForestClassifier
```

XGBoost Classifier	The (params) define a grid for hyperparameter tuning of the XGBoost Classifier (XGBClassifier), including min_child_weight, gamma, colsample_bytree, and max_depth. The XGBClassifier is configured with a learning rate of 0.5, 100 estimators, using a binary logistic regression objective, and utilizing 3 threads for processing. GridSearchCV (xg_cv) is used with 5-fold cross-validation (cv=5), refitting the best model (refit=True), evaluating based on accuracy (scoring="accuracy") <pre> from sklearn.model_selection import GridSearchCV from sklearn import xgbclassifier rf = xgbclassifier() param_grid = {'n_estimators': [100, 200, 300], 'learning_rate': [0.1, 0.2], 'max_depth': [10, 20, 30], 'min_child_weight': [1, 2, 3]} grid_search = GridSearchCV(rf, param_grid, cv=5, n_jobs=-1, verbose=1) grid_search.fit(X_train, y_train) # fit Fitting 5 folds for each of 16 candidates, totaling 80 fits. = GridSearchCV (0.0) = best_estimator_: xgbclassifier xgbclassifier 1 </pre>
---------------------------	---

Final Model Selection Justification (2 Marks):

Final Model	Reasoning
-------------	-----------

Random Forest

Random Forest model is chosen for its robustness in handling complex datasets and its ability to mitigate overfitting while providing high predictive accuracy.



Above all the models Random Forest model have the highest accuracy among all the models.